

Sameer Karam

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EDUCATION

McGill University

Bachelor of Computer Engineering

Aug 2021 – May 2025

Montreal, Canada

Audacity Launchpad (sponsored by Google)

Nanodegree in AI Programming with Python and TensorFlow

Aug 2025 – Present

Remote

EXPERIENCE

Software Engineer Intern

Jaffa.Net

Jul 2023 – Sept 2023

Remote

- **Optimized backend order-processing pipelines in Python and Java**, improving transaction throughput by **25%**.
- **Integrated external REST APIs** with the restaurant platform to synchronize inventory and menu data with the database layer.
- Implemented monitoring and logging to surface latency bottlenecks, reducing average response time across high-traffic endpoints.
- Contributed to Agile sprints, code reviews, and CI, improving feature delivery cadence by **20%**.

PROJECTS

AI-Powered Career Guidance Platform | *FastAPI, Python, React, GPT-4, PostgreSQL*

Jul 2025 – Present

- **Built Python data pipelines** ingesting **BLS** and **FRED** APIs for real-time labor-market insights.
- **Applied prompt-engineering**: structured user profile data + economic indicators into **GPT-4** prompts to generate personalized, market-aligned career pathways.
- Developed **FastAPI** backend with **SQLAlchemy ORM**, auth, and REST endpoints powering an interactive React dashboard.

Full-Stack High-Speed Rail Feasibility Simulator | *FastAPI, Python, React*

Sept 2024 – Apr 2025

- **Extracted and processed** Eurocontrol flight lists and Deutsche Bahn datasets to simulate **flight-to-rail** replacements.
- Designed APIs exposing **CO₂**, time, and cost comparisons across **200+** routes; visualized insights via **Leaflet/Plotly**.
- Implemented clean, reusable modules (*framework over features*) to support fast iteration on new routes and metrics.

Tumor Segmentation and Detection in Medical Imaging | *Python, OpenCV, scikit-learn*

Jan 2025 – Apr 2025

- Applied K-means and SVM segmentation on MRI datasets, achieving a **28%** improvement in DICE scores.
- Hardened the pipeline with preprocessing (denoising, augmentation) and validation to improve robustness.

Automotive Chatbot (Matador.ai Hackathon) | *Python, NLP*

Nov 2024

- Built a **spaCy**-based assistant for vehicle recommendations using intent/rule matching; ranked in the top **85%**.

Personal Portfolio Website | *HTML, CSS, JavaScript*

- Designed and deployed a personal website to showcase technical projects and resume.
- More projects available at: samraa66.github.io/My-Portfolio-Website

SKILLS

- **Programming Languages:** Python, SQL, Java, C, C#, ARM Assembly
- **Tools:** Git, Docker, Excel, VS Code
- **Languages:** Arabic (Fluent), English (Fluent), French (Basic working proficiency)